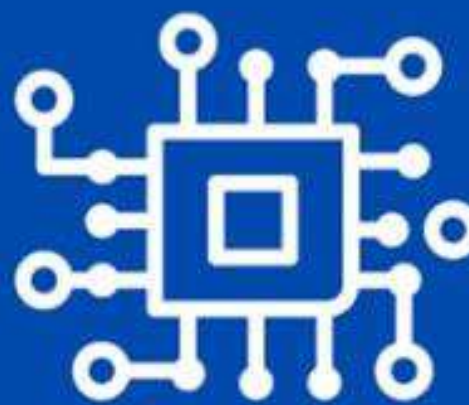




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NUMERICAL MODELING AND OPTIMIZATION OF GROUNDWATER FILTRATION PROCESSES USING COMPUTATIONAL ALGORITHMS

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Annotation. This study develops a reproducible computational framework for groundwater flow simulation and constrained abstraction planning. A conservative finite-volume scheme with harmonic interface transmissivity and backward Euler integration represents transient flow in a heterogeneous confined aquifer. Linear response coefficients connect the numerical model to a convex quadratic allocation problem. A controlled 90-day experiment distributes a demand of 1,200 m³/day among three abstraction zones. Compared with equal allocation, the nominal optimum reduces mean squared drawdown by 25.87% and maximum drawdown by 30.88%, while preserving the total supply. A 20% conductivity reduction exposes a small violation of the hydraulic-head constraint; explicit scenario constraints remove this violation. Additional experiments quantify the effects of a prescribed head buffer and the maximum feasible abstraction under the stated boundaries. Analytical benchmarks, mesh refinement, an independent linear solver and water-balance checks support numerical consistency. The results demonstrate the computational mechanism under synthetic conditions and define the field information needed before practical application.

Keywords: groundwater flow; filtration; finite-volume method; harmonic transmissivity; backward Euler method; response matrix; quadratic programming; uncertainty analysis.

HISOBLASH ALGORITMLARI YORDAMIDA YER OSTI SUVLARI FILTRATSIYASI JARAYONLARINI SONLI MODELLASHTIRISH VA OPTIMALLASHTIRISH

Botirov Munisbek

Magistrant, Qarshi davlat texnika universiteti

Qarshi, O'zbekiston

Annotatsiya. Maqolada yer osti suvlari oqimini sonli modellashtirish va suv olishni cheklovlar asosida rejalashtirish uchun takroran hisoblash mumkin bo'lgan yondashuv ishlab chiqilgan. Geterogen bosimli suvli qatlamdagi o'tkinchi oqim chegaralarda garmonik transmissivlikni qo'llovchi konservativ chekli hajmlar sxemasi va teskari Eyler usuli bilan ifodalanadi. Chiziqli ta'sir koeffitsiyentlari sonli modelni qavariq kvadratik optimallashtirish masalasi bilan bog'laydi. 90 sutkalik sintetik tajribada 1 200 m³/sutka talab uchta suv olish hududi o'rtasida taqsimlanadi. Teng taqsimotga nisbatan nominal optimal yechim napor pasayishining o'rtacha kvadratini 25,87%, eng katta pasayishni esa 30,88% kamaytiradi. Gidravlik o'tkazuvchanlik 20% pasayganda yuzaga keladigan cheklov buzilishi ssenariylarni hisobga olish orqali bartaraf etiladi. Napor zaxirasi va suv olishning hisobiy chegarasi alohida baholanadi. Analitik sinovlar, to'rtinchi zichlashtirish, mustaqil yechuvchi va suv balansi tekshiruvlari sonli izchillikni tasdiqlaydi. Natijalar haqiqiy dala kuzatuvlari o'rnini bosmaydi.

Kalit so'zlar: yer osti suvlari oqimi; filtratsiya; chekli hajmlar usuli; garmonik transmissivlik; teskari Eyler usuli; ta'sir matritsasi; kvadratik dasturlash; noaniqlik tahlili.

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INTRODUCTION

Groundwater abstraction changes both hydraulic heads and the pathways by which an aquifer receives water. A plan that satisfies a volumetric demand can therefore remain hydraulically undesirable: concentrating abstraction in a poorly conductive zone may produce excessive local drawdown even when sufficient water enters elsewhere. Numerical modeling becomes useful when it connects these spatial effects to explicit decisions about where and how much water to withdraw. In this article, “filtration” denotes seepage through a saturated porous medium; contaminant removal and solute transport are outside the model.

The credibility of such a calculation depends on more than obtaining a converged solution. The conceptual aquifer, boundary conditions, discretization and decision constraints must describe the same problem. Reilly and Harbaugh [1] emphasize evaluation of the model in relation to its intended use. The MODFLOW 6 documentation [2] provides a broader reference for groundwater-flow formulations and numerical implementation. A small, transparent model is useful for examining the connection between conservation, optimization and uncertainty before adopting a more detailed spatial representation.

The aim is to develop and evaluate a computational framework that minimizes hydraulic disturbance while meeting a prescribed supply requirement. The study examines the discretization of flow across a sharp conductivity contrast, the reduction of repeated flow simulations to an allocation problem, and the effect of conductivity uncertainty on feasibility. A secondary calculation distinguishes the preferred allocation at a fixed demand from the maximum demand permitted by the model.

The contribution is an auditable numerical experiment combining interface treatment, transient response coefficients, constrained allocation and verification. The underlying methods are established. The added value lies in demonstrating their interaction, including a nominally feasible plan that fails under a modest parameter change and the cost of correcting that failure. All reported values are generated from stated synthetic inputs. No field calibration, observed aquifer response or economic benefit is claimed.

LITERATURE REVIEW

Groundwater optimization includes several distinct tasks. Parameter estimation adjusts aquifer properties to observations; experimental design selects informative measurements; management optimization selects actions under a calibrated or assumed model. Yeh [3] reviews these roles and the use of linear, quadratic, nonlinear and stochastic programming. The present work concerns management optimization with prescribed parameters. Its conductivity scenarios test sensitivity and feasibility; they do not constitute an inverse estimate of hydraulic conductivity.

Norouzi Khatiri et al. [4] organize groundwater management research around simulation, optimization, uncertainty analysis and decision-making. Their synthesis supports treating uncertainty as part of the decision formulation rather than only reporting it after selecting a plan. Here, the uncertainty set is deliberately small and explicit. It represents three conductivity multipliers, without assigning probabilities or claiming coverage of all possible geological conditions.

Conservation is central to the numerical layer. Finite-volume methods express the governing equation as a balance over each control volume, with shared fluxes canceling at internal faces [5]. In a layered one-dimensional medium, harmonic averaging follows from adding hydraulic resistances in series. This argument does not justify the same two-point

approximation for every anisotropic or irregular mesh. Provost et al. [6] show why connectivity and geometric representation matter for flow through dipping aquifers in MODFLOW 6. The benchmark below uses aligned interfaces, scalar conductivity and a horizontal strip, where the simpler treatment is appropriate.

Response coefficients provide a bridge between a flow model and an optimizer. The USGS Ground-Water Management process [7] documents this approach for relating controlled stresses to hydraulic responses. With fixed transmissivity and linear boundary conditions, superposition can be exact for the discrete model. If transmissivity changes with saturated thickness, or boundary behavior switches as heads change, the coefficients generally require updating, and the optimization loses its simple linear-response structure.

Quadratic programming is a natural choice when the disturbance measure is a sum of squared drawdowns and constraints are affine. Standard optimization theory [8] then provides stronger diagnostics than a solver success message alone: curvature, feasibility and optimality conditions can be checked separately. This study uses scientific Python [9] for matrix operations and optimization, but verifies the implemented flow solver independently. The literature therefore motivates a focused workflow in which physical assumptions, numerical error and decision sensitivity remain visible throughout the calculation.

RESEARCH METHODOLOGY

Conceptual aquifer and experimental design

The model represents a confined aquifer as a horizontal strip of length 1,000 m, width 1,000 m and thickness 20 m. Flow is averaged across the width and thickness. Three 100 m abstraction intervals represent distributed extraction zones, rather than individual wells with radial flow or well losses. A central low-conductivity layer occupies 400–600 m. The aquifer roof is assigned elevation 80 m; all computed heads remain above it, so constant saturated thickness is internally consistent.

The left and right boundaries maintain heads of 100 and 98 m, respectively. A uniform distributed source represents effective recharge or leakage into the modeled layer, without modeling an unsaturated zone. The initial condition is the steady solution with this source and zero abstraction. At time zero, constant extraction begins and continues for 90 days. Table 1 lists the inputs; none was fitted to observations.

Table 1.

Inputs to the controlled groundwater experiment

Parameter	Value	Interpretation
Geometry	1,000 × 1,000 × 20 m	Length × width × thickness
Hydraulic conductivity, K	10; 2; 10 m/day	Intervals 0–400; 400–600; 600–1,000 m
Storativity, S	0.002	Dimensionless confined storage
Distributed source, r	0.0002 m/day	Total input: 200 m ³ /day
Boundary heads	100; 98 m	Left; right, fixed in time
Abstraction intervals	200–300; 450–550; 700–800 m	Zones 1, 2 and 3

Parameter	Value	Interpretation
Demand and zone bounds	1,200; 100–650 m ³ /day	Total demand; bound per zone
Minimum permitted head	97.50 m	Applied at every cell center
Base discretization	$\Delta x = 10$ m; $\Delta t = 0.5$ day	100 cells; 180 time steps
Conductivity multipliers	0.8; 1.0; 1.2	Applied to the entire K field

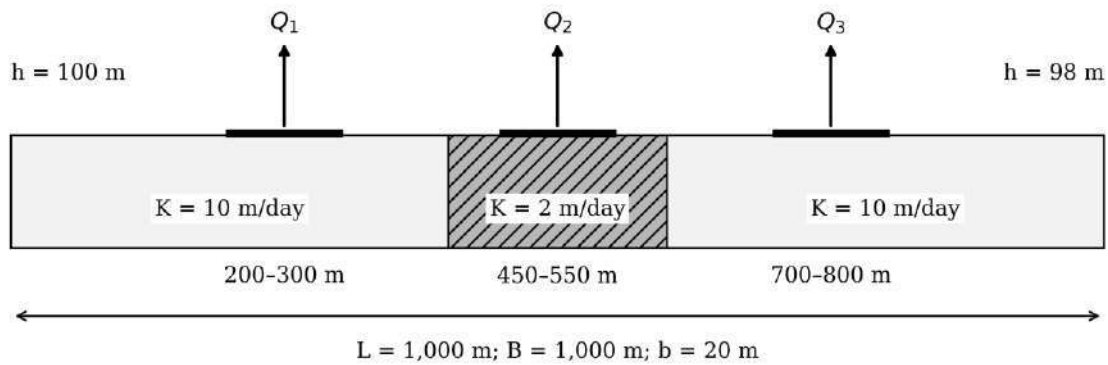


Figure 1. Aquifer geometry and abstraction zones

Governing equation and conservative discretization

Let h be hydraulic head, $T = Kb$ transmissivity, B aquifer width, and Q_j the volumetric abstraction rate from interval I_j of length ℓ_j . With positive Q_j denoting removal, the governing equation is

$$S \frac{\partial h}{\partial t} = \frac{\partial}{\partial x} \left(T \frac{\partial h}{\partial x} \right) + r - \sum_{j=1}^3 \frac{Q_j \chi_{I_j}(x)}{B \ell_j} \quad (1)$$

Here χ is the interval indicator. Each term has units of length per time. The boundary and initial conditions are

$$h(0, t) = 100, \quad h(L, t) = 98, \quad h(x, 0) = h^0(x) \quad (2)$$

The steady unpumped solution h^0 satisfies Equation (1) with zero time derivative and $Q_j = 0$. Cell-centered finite volumes are used. At an interior face between equal-width cells, transmissivity is

$$T_{i+1/2} = \frac{2T_i T_{i+1}}{T_i + T_{i+1}} \quad (3)$$

Equation (3) preserves the series resistance of the two half-cells. At an external boundary, the head-to-cell-center distance is $\Delta x/2$, giving a boundary coefficient $2T_i/\Delta x^2$. Abstraction is distributed by the actual overlap between each cell and each interval, which preserves the assigned Q_j even if a zone boundary does not coincide with a cell face.

For an interior cell, backward Euler integration gives

$$-a_i h_{i-1}^{n+1} + \left(\frac{S}{\Delta t} + a_i + c_i \right) h_i^{n+1} - c_i h_{i+1}^{n+1} = \frac{S}{\Delta t} h_i^n + r - w_i \quad (4)$$

The coefficients are $a_i = T_{i-1/2}/\Delta x^2$ and $c_i = T_{i+1/2}/\Delta x^2$; w_i is the cell-averaged removal per unit area. Boundary contributions move to the right-hand side. Positive storage and transmissivity give a strictly diagonally dominant tridiagonal matrix with nonpositive off-

diagonal entries. The Thomas algorithm therefore provides an efficient direct solution with linear work per time step. Backward Euler avoids an explicit diffusion stability restriction, although time-step refinement remains necessary for accuracy.

Response matrix and allocation objective

Drawdown is measured relative to the unpumped equilibrium: $s = h^0 - h$. Define a fixed reference rate $D_r = 1,200 \text{ m}^3/\text{day}$ and dimensionless controls $p_j = Q_j/D_r$. For each zone, one response simulation applies D_r to that zone and zero extraction to the others. These basis runs are mathematical probes and need not satisfy the operational upper bounds. Let A^n contain the three resulting drawdown vectors at time level n . Linearity gives

$$\mathbf{s}^n = A^n \mathbf{p}, \quad \mathbf{h}^n = \mathbf{h}^0 - A^n \mathbf{p} \quad (5)$$

The objective is mean squared drawdown over all N cells and N_t time levels:

$$J(\mathbf{p}) = \frac{1}{NN_t} \sum_{n=1}^{N_t} \|A^n \mathbf{p}\|_2^2 = \mathbf{p}^T H \mathbf{p} \quad (6)$$

$$H = \frac{1}{NN_t} \sum_{n=1}^{N_t} (A^n)^T A^n, \quad \nabla J = 2H\mathbf{p} \quad (7)$$

J has units of m^2 . It measures hydraulic disturbance, not pumping energy, financial cost or depletion of a finite reserve. Uniform space and time steps make the arithmetic average consistent with a space-time average. Nonuniform discretizations would require appropriate weights.

For a demand D , the feasible controls satisfy

$$\sum_{j=1}^3 p_j = \frac{D}{D_r}, \quad \frac{100}{D_r} \leq p_j \leq \frac{650}{D_r} \quad (8)$$

The lower bounds are assumed service obligations that keep all zones active. They are decision inputs, not hydraulic laws. The nominal model additionally requires $h_i^n \geq 97.50 \text{ m}$. The optimizer uses sequential least-squares programming (SLSQP), an analytical gradient, a stopping tolerance of 10^{-12} and a maximum of 500 iterations. Because H is a Gram matrix, the objective is convex. Positive eigenvalues in the experiment establish strict convexity and uniqueness of a feasible minimizer.

Conductivity scenarios, head buffers and feasibility

The three conductivity scenarios use K multiplied by 0.8, 1.0 and 1.2. Each has its own unpumped equilibrium and response matrix. The scenario-constrained problem minimizes nominal J while imposing

$$\mathbf{h}^{0,\omega} - A^{N_t,\omega} \mathbf{p} \geq (97.50 + \delta) \mathbf{1}, \quad \omega \in \{0.8, 1.0, 1.2\} \quad (9)$$

The optional buffer δ is a prescribed additional head margin, in meters. It is not a statistical confidence interval. Under constant nonnegative extraction, an equilibrium start and the monotone linear scheme, drawdown increases toward its steady value. The final-time head is therefore the limiting head for each cell. All intermediate time levels are also checked after optimization. This reduction is not generally valid for seasonal pumping, changing recharge or switching boundaries.

A separate linear program maximizes total abstraction under the same zone bounds and head constraints:

$$D_{\max} = D_r \max_{\mathbf{p}} \mathbf{1}^T \mathbf{p} \quad (10)$$

The capacity problem uses the HiGHS linear programming solver through SciPy. This separates infeasibility from a failure of the nonlinear optimization routine. The resulting maximum is conditional on the model, horizon and boundary heads; it is not an estimate of sustainable aquifer yield.

Verification and computational workflow

Verification uses four complementary checks. First, a homogeneous diffusion mode is compared with the analytical solution

$$u(x, t) = u_0 \sin\left(\frac{\pi x}{L}\right) \exp\left(-\frac{T \pi^2 t}{SL^2}\right) \quad (11)$$

The benchmark has zero endpoint values, $T = 200 \text{ m}^2/\text{day}$, $S = 0.002$, $L = 1,000 \text{ m}$, and initial amplitude $u_0 = 1 \text{ m}$. Errors are evaluated at one day. Spatial error is isolated by exact integration of the semidiscrete system, while temporal error is measured against that system's exact solution. Second, a layered steady-flow test with no source or extraction checks interface fluxes. Third, SciPy's independent banded solver recomputes the optimized transient heads directly and compares them with Equation (5).

Finally, every time step is checked against the global balance

$$Q_L - Q_R + rBL - \sum_j Q_j = SB\Delta x \sum_i \frac{h_i^{n+1} - h_i^n}{\Delta t} \quad (12)$$

Q_L is positive inward at the left boundary, and Q_R is positive outward at the right boundary. The relative residual is the absolute imbalance divided by the sum of the absolute magnitudes of the five balance terms. This normalization remains meaningful when a boundary flux reverses.

The implementation uses Python, NumPy 2.3.5 and SciPy 1.17.0 in double precision. A run assembles the grid and overlaps, solves the initial equilibrium, generates response columns, forms H , solves the allocation problem, and recomputes heads and balances. The workflow is deterministic and uses no random sample. The three response columns can be propagated together. For m controls, response generation requires $O(mNN_t)$ work; evaluating the reduced objective subsequently depends on the small $m \times m$ matrix rather than another transient solve. For many controls, adjoint or reduced-order approaches would deserve comparison.

ANALYSIS AND RESULTS

Numerical accuracy and interface behavior

Table 2 reports representative verification results. The smooth benchmark recovers approximately second-order spatial convergence and first-order temporal convergence. These orders describe the benchmark, not a universal error bound for heterogeneous field models. Refining the operational grid from 10 to 5 m changes the optimized J by 0.0296%; reducing the time step from 0.5 to 0.25 day changes it by 0.0508%. Thus, the reported allocation benefit is much larger than these discretization changes.

Table 2.

Numerical verification and refinement diagnostics

Check	Setting	Result
Spatial benchmark	$\Delta x: 20 \rightarrow 10 \text{ m}$	RMS error: $8.5575 \times 10^{-5} \rightarrow 2.1393 \times 10^{-5} \text{ m}$; order ≈ 2.00
Temporal benchmark	$\Delta t: 0.025 \rightarrow 0.0125 \text{ day}$	RMS error: $0.003176 \rightarrow 0.001596 \text{ m}$; order ≈ 0.99
Layered steady flux	$\Delta x = 10 \text{ m}$; harmonic mean	$222.2222 \text{ m}^3/\text{day}$; maximum head error $< 2 \times 10^{-12} \text{ m}$

Check	Setting	Result
Layered steady flux	$\Delta x = 10$ m; arithmetic mean	225.5639 m ³ /day; flux overestimate: 1.5038%
Independent transient solve	Nominal optimized policy	Maximum head difference: 1.80×10^{-11} m
Global water balance	All reported grid refinements	Maximum relative residual $< 3 \times 10^{-10}\%$

For the layered steady test, the exact flow is 222.2222 m³/day, obtained by dividing $B(100 - 98)$ by the total hydraulic resistance $400/200 + 200/40 + 400/200$. Harmonic averaging reproduces this resistance to numerical precision. Arithmetic averaging overestimates flow by 6.30% on a 40 m grid and 1.50% on a 10 m grid. This difference illustrates why a small algebraic residual alone cannot verify interface physics: either discretization can solve its own equations accurately.

Allocation performance and physical interpretation

The equal-allocation benchmark withdraws 400 m³/day from each zone. The nominal optimum assigns 550, 100 and 550 m³/day, respectively (Table 3). The central zone reaches its lower bound because abstraction within the low-conductivity layer produces greater drawdown. The outer allocations are equal because the drawdown-response system is symmetric, even though the absolute boundary heads differ.

The nominal optimum reduces J by 25.87% and maximum drawdown by 30.88%. Both plans deliver 108,000 m³ over 90 days. The result is therefore an improvement in the spatial distribution of hydraulic disturbance, with no reduction in delivered volume. The minimum absolute head improves by only 0.0190 m: minimizing squared drawdown is not equivalent to maximizing the lowest head because the unpumped head varies spatially.

Table 3.

Nominal results at $D = 1,200$ m³/day

Policy	$Q_1 / Q_2 / Q_3$, m ³ /day	J , m ²	Maximum drawdown, m	Minimum head, m
Equal allocation	400 / 400 / 400	0.543822	1.275000	97.635567
Nominal optimum	550 / 100 / 550	0.403149	0.881250	97.654583
Scenario-constrained, $\delta = 0$	555.041 / 100 / 544.959	0.403158	0.881425	97.658847

The eigenvalues of H are approximately 0.07075, 0.25225 and 1.76193 m². The two free-control objective gradients agree within 1.4×10^{-15} , while the active lower-bound multiplier for the central control is positive, approximately 0.48724. Together with feasibility, these conditions support optimality of the nominal solution. SLSQP terminates in two iterations; this small count reflects the low-dimensional convex problem and should not be extrapolated to

regional nonlinear models.

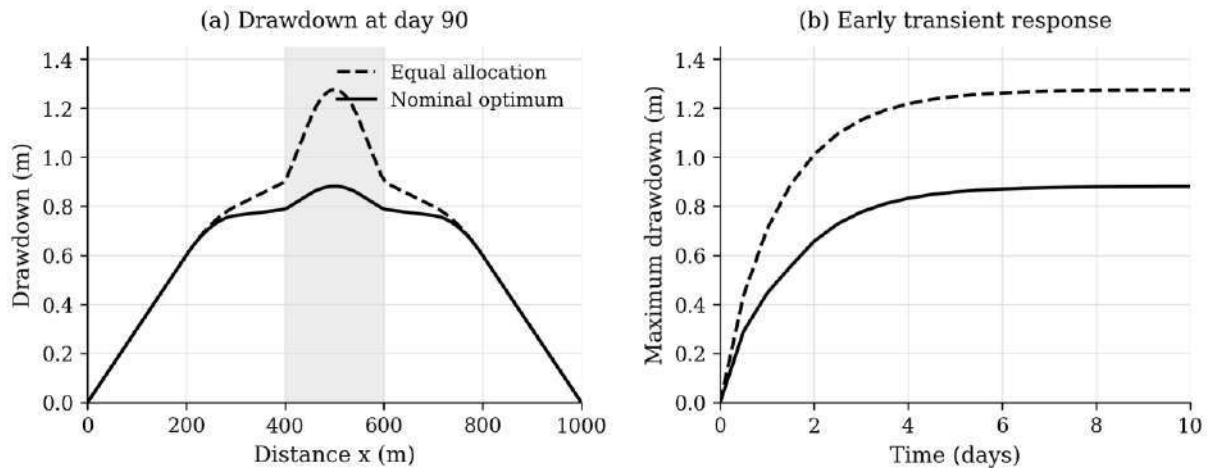


Figure 2. Final drawdown distribution and early transient response

Under the nominal optimum, maximum drawdown is 0.4451 m after one day, 0.8586 m after five days and 0.8808 m after ten days. It is effectively steady by day 30. The long horizon consequently gives substantial weight to the near-steady regime. A short emergency-pumping horizon could produce a different objective and should be evaluated separately.

At the final nominal state, the left boundary supplies 722.222 m³/day and the right boundary supplies 277.778 m³/day; distributed input supplies the remaining 200 m³/day. The right flux is therefore inward, despite being defined as positive outward in Equation (12). Storage release matters during the transient but becomes negligible at equilibrium. Fixed-head boundaries act as idealized sources that maintain their prescribed heads, an assumption central to interpreting the available supply.

Scenario protection and the value of an explicit buffer

Applying the nominal optimum to the 0.8K scenario gives a minimum head of 97.494632 m, violating the constraint by about 0.00537 m. The scenario-constrained solution shifts approximately 5.041 m³/day from the right zone to the left and restores the limiting head to 97.50 m. Its nominal J increases by only 0.0022%. The unequal boundary heads now matter through the absolute-head constraint, breaking the symmetry of the allocation.

Table 4.

Head buffers and scenario-constrained allocation

Buffer δ , m	$Q_1 / Q_2 / Q_3$, m ³ /day	Nominal J , m ²	Lowest scenario head, m
0.00	555.041 / 100 / 544.959	0.403158	97.50
0.02	573.821 / 100 / 526.179	0.403348	97.52
0.05	601.992 / 100 / 498.008	0.404097	97.55
0.10	648.944 / 100 / 451.056	0.406579	97.60

Meeting a constraint exactly leaves no additional model margin. Table 4 therefore examines δ from zero to 0.10 m. A 0.05 m buffer increases nominal J by approximately 0.235% relative to the nominal optimum; a 0.10 m buffer increases it by approximately 0.851%. The latter almost exhausts the left zone's 650 m³/day upper bound. A prescribed buffer can make a policy less sensitive to small discrepancies, but its size must ultimately be tied to observation

error, model discrepancy and management requirements.

Demand limits, transferability and limitations

Increasing total demand changes which constraints control the solution. At 800 and 1,000 m³/day, equal outer-zone rates with the central zone at its minimum satisfy all scenarios. At 1,300 m³/day, the left zone reaches its upper bound and the central rate rises to 164.379 m³/day. The maximum feasible demand is 1,540.64 m³/day under nominal conductivity and 1,349.53 m³/day when all three scenarios are enforced. The latter is 12.40% lower. These values come from the separate linear program, rather than treating an SLSQP failure as proof of infeasibility.

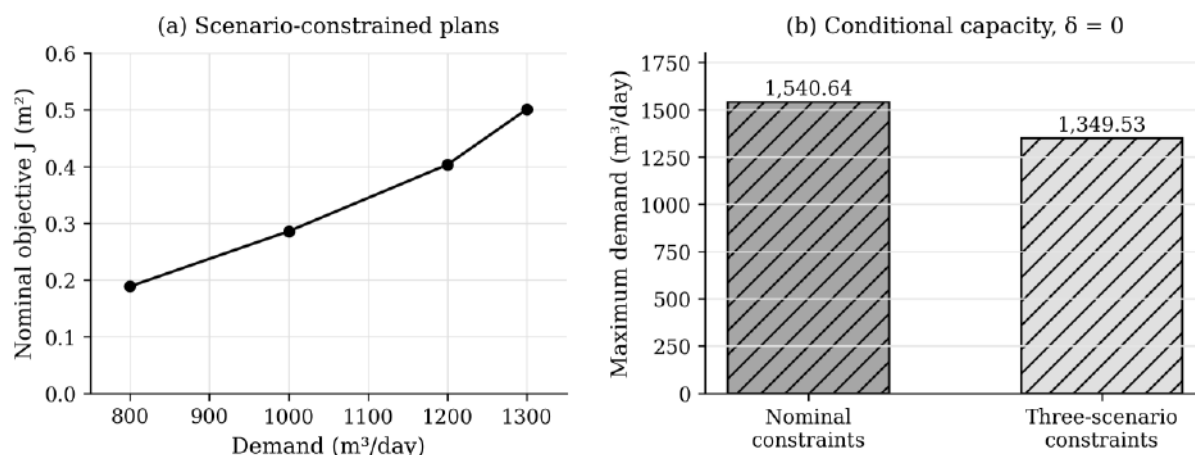


Figure 3. Demand sensitivity and conditional abstraction limits

A requested demand of 1,400 m³/day is infeasible under the scenario constraints and zone bounds. The correct response is to revise supply requirements, bounds or available sources after a physical assessment, rather than silently relaxing the head criterion. The gap between nominal and scenario-constrained capacity demonstrates why optimizing a single assumed conductivity field can overstate operational flexibility.

Several limitations govern transferability. Width averaging cannot resolve individual well interference, near-well gradients or partial penetration. Uniform conductivity multipliers omit uncertain layer geometry, spatially correlated properties, recharge variability and boundary uncertainty. The model also omits unconfined drainage, stream depletion limits, subsidence, water quality and pumping energy. Scenario feasibility provides no probability of success outside the three specified parameter fields.

Practical application would require geological logs, pumping and recovery tests, time-resolved abstraction, observation-well heads and evidence supporting the boundary conditions. Calibration should distinguish parameters that observations can constrain from those that remain uncertain, and evaluation should use withheld periods or independent measurements. A two- or three-dimensional flow model could then replace the strip model. The response-matrix interface remains useful only where linearity is defensible; otherwise, sequential relinearization and independent forward checks are necessary.

CONCLUSION AND RECOMMENDATIONS

A conservative flow model and a response-based allocation problem have been integrated into a reproducible groundwater-planning experiment. Harmonic interface transmissivity preserves layered hydraulic resistance, while implicit time integration and a tridiagonal solver provide a stable computational core. Analytical and independent-solver

checks support the numerical implementation, and explicit water balances make the sources of extracted water traceable.

For the prescribed 1,200 m³/day demand, the nominal allocation of 550/100/550 m³/day reduces mean squared drawdown by 25.87% and maximum drawdown by 30.88% relative to equal allocation. Conductivity uncertainty changes feasibility: a 20% reduction produces a small head violation, removed by the scenario-constrained allocation of approximately 555.041/100/544.959 m³/day. Additional head buffers expose a clear trade-off between average disturbance and protection of the lowest head.

Before adopting an allocation, verify interface fluxes and water balance, report supply and hydraulic responses separately, and test parameter scenarios with explicit operating margins. Field use requires confirmation of the assumed aquifer properties and boundary conditions.

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